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Perspective

Chemical Synthesis of Adipic Acid from Glucose and Derivatives: Challenges for Nanocatalyst Design

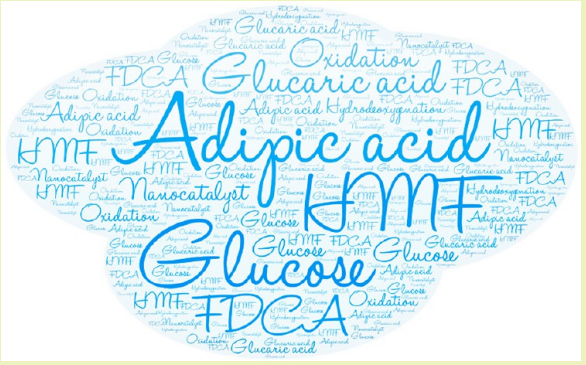
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ABSTRACT: Adipic acid is an essential building block for the nylon industry, but its production continues to rely on fossil-derived benzene and generates a substantial amount of N_2O every year. Recognizing the environmental impact of conventional adipic acid production and motivated by societal interests toward a sustainable future, the production of adipic acid should switch toward a renewable feedstock and proceed with a nitric acid-free protocol. In light of this transition, glucose emerges as a suitable feedstock to replace benzene for bioadipic acid production. It can be chemically converted into bioadipic acid *via* two different routes, forming either glucaric acid or 2,5-furandicarboxylic acid as the respective key intermediates. However, these transformations are challenged by various issues, such as retro-aldol condensation, which fragments glucose into smaller organic byproducts, and catalyst instability, which compromises the reaction rates. This perspective addresses the potentials and challenges in nanocatalyst design and discusses how catalyst morphology tailoring and metal−support interaction tuning can be revised systematically to maximize the yield of bioadipic acid. The influence of catalyst structures on selective activation of C−H (oxidation) and C−O bond (hydrogenolysis) of the intermediates is critically discussed. The electronic features of bimetallic and metal−acid/base hybridized catalysts on C−H and C−O bond activation have been correlated in this work. The goal of this perspective aims to provide insights on novel nanocatalysts designed to improve bioadipic acid production from sustainable feedstocks.



KEYWORDS: *Adipic acid, Glucose, Nanocatalyst, Oxidation, Hydrodeoxygenation bioadipic acid, Environmentally beneficial chemical, Enhanced global sustainability*

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INTRODUCTION

Adipic acid is recognized globally as one of the most critical commodity chemicals for its versatile applications to produce many important chemicals, such as Nylon-6,6, polyurethanes, cosmetics, gelatins, lubricants, fertilizers, adhesives, and insecticides.[1−3] Global production for adipic acid is projected to reach over 6.9 billion pounds by 2021,[4−6] and notably, the production of Nylon-6,6 accounts for almost 75% of the total output.[7,8]

Fossil-Derived Adipic Acid. Traditional industrial production of adipic acid begins with the hydrogenation of petroleum-derived benzene using a Ni−Al_2O_3 catalyst to yield cyclohexane.[9] Subsequently, cyclohexane is oxidized in a liquid phase to a mixture of cyclohexanone and cyclohexanol, commonly referred as KA oil, in air with a catalytic load of cobalt naphthenate. The KA oil is then reacted with a stoichiometric amount of nitric acid in the presence of a Cu−V-based catalyst to achieve ring-opening reactions to yield adipic acid (Figure 1, orange). The adipic acid is then purified through steam distillation and recrystallization.[10] However, conversion of cyclohexane is controlled below 10% to avoid an explosive envelop, implying that a considerable amount of energy is lost for product separation and recycling processes.[11] The use of nitric acid, however, leads to the generation of N_2O, which has a global warming potential (GWP) almost 300 times greater than that of CO_2. Despite the use of sophisticated recovery and regenerative technologies, an estimated amount of 400,000 t of N_2O still leaks into the atmosphere, contributing 5%−8% of the global N_2O generation every year.[12]

Adipic acid can also be synthesized from butane, which begins with a dehydrogenation to yield 1,3-butadiene, followed by hydroformylation at both ends to yield adipaldehyde. This is partly due to the high reactivity of the aldehyde groups which

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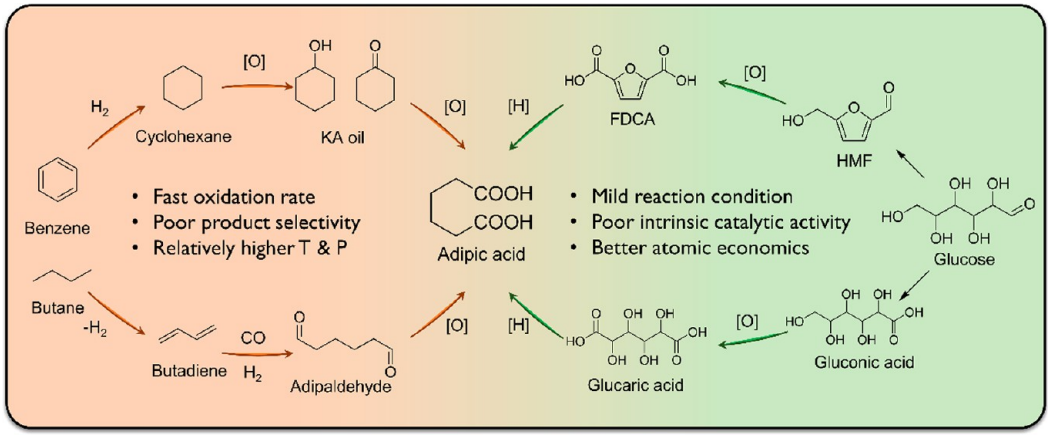


Figure 1. Adipic acid synthesis from conventional fossil feedstock and biomass feedstocks covered in this perspective.

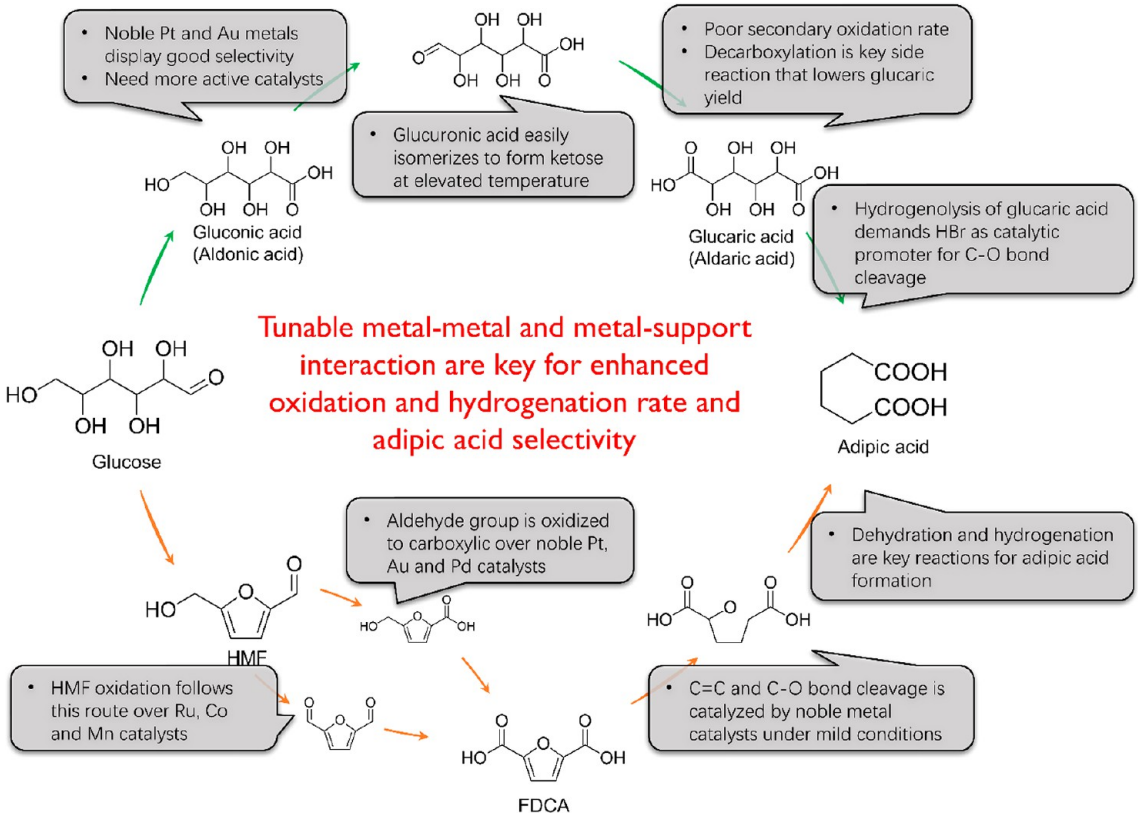


Figure 2. Two possible routes of chemical synthesis of adipic acid from glucose.

undergoes aldol before the adipaldehyde is isolated. In 1994, Union Carbide patented a Rh complex-catalyzed protocol for an *in situ* acetalization to reach a product selectivity at 57%.[13] Later, Smith and co-workers reported another Rh-catalyzed protocol and furthered the selectivity to 74% for adipaldehyde, which can be oxidized to adipic acid using oxone.[14] Liu and co-workers reported cost-effective hydroformylation processes, which show significant environmental and economic promises for industrial implementation.[15−18] Overall, the protocol of the butadiene route offers a nitric-acid free route to access adipic acid, but its yield and efficiency requires significant improvement compared to the convention cyclohexane route. Obviously, there is a strong need to develop alternative oxidants for this process.

Extensive research efforts have been devoted to eliminate the use of nitric acid to oxidize cyclohexane to adipic acid. Alternative oxidants such as air and H_2O_2 have been proposed and investigated over homogeneous and heterogeneous catalysts. Supported Co, Fe, Au, Pd, Pt, Ru, Ag on TiO_2, SiO_2, CeO_2, MgO, ZrO_2, WO_3, SiC, and Al_2O_3 have been reported for efficient oxidation under 150 °C in the presence of air.[19−22] Atomically dispersed single-site catalysts, including tungstic acid,[23−28] $H_3PW_4O_{24}$,[29] Mo complexes,[30] Mn porphyrins,[31] and Fe- and Ti-phosphate,[32,33] have also been found to be effective for adipic acid synthesis from KA oil.[34,35] In addition, Sato and co-workers[36] reported direct oxidation of cyclohexene to adipic acid using of 30% H_2O_2 as oxidant catalyzed by Na_2WO_4. These advances improve the production process by eliminating the emission of N_2O, but they still rely on petroleum-derived benzene, which challenges the notion of sustainability.

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Bioderived Adipic Acid. *Background.* The transition from petroleum to biomass feedstocks for adipic acid production has been explored by many research groups. These efforts are motivated by the renewable aspect of biobased feedstocks, and their potential to reduce business risk and environmental impact, compared to petroleum resources.[4,6,20,35,36] The application of the downstream products, price, and abundance of biobased precursors and physiochemical properties of resultant polymers, as well as potential wastes generated during manufacture of bioadipic acid, are all critical factors to be considered for future business models. Life cycle analysis showed that biobased adipic acid produced from a lignin waste stream from the bioethanol refinery can lead to a reduction from 62% to 78% of GHG emission, compared to conventional production routes that use nitric acid.[37]

Currently, there are three catalytic routes to produce bioadipic acid: biocatalysis, chemocatalysis, and the combination of both. High production cost is still the main barrier to commercializing bioadipic acid,[38] particularly with expensive noble metals and poor chemoselectivity toward adipic acid and imtermediates.[5,39] The current mainstream bioproduction of bioadipic acid from glucose is achieved via biocatalysis,[8,40,41] using a reverse β-oxidation pathway in *E. coli* under anaerobic conditions. The advances on biocatalysis have been reviewed recently,[38,42−47] which covered the biological pathway and the challenges to transform polysaccrides, lignin, and fatty acids to adipic acid. Those reviews also concluded that the biocatalysis pathway using natural occurring microorganisms is yet to be efficient.

Despite the low reaction rates, biosynthesis is still producing a greater yield than chemocatalysis,[48−50] as glucose is susceptible to commence retro-aldol condensation at elevated temperatures, yielding smaller organic fragments instead of adipic acid. Nevertheless, it is worth noting that chemocatalysis has several advantages over biocatalysis. First, chemocatalysis can eventually benefit from the existing petroleum refinery infrastructures and chemical supply chains, both of which can provide an economic advantage for bioadipic acid. For example, process intensification technologies used in oxidation of hydrocarbons can be readily available for oxidation of sugars.[51,52] However, for most biological applications, complicated configuration of vessels and unit operations for storage of enzymes is inevitable, thus increasing the overall capital cost. Second, a possible intermediate during the chemical conversion of glucose to bioadipic acid is FDCA.[53] Thus, the practice of chemocatalysis for bioadipic acid production could promote biorefinery practices by employing well-established platform chemicals.

Notably, Rennovia has recently commercialized the production of biobased adipic acid, and the reaction proceeded using air as an oxidant. The process is catalyzed heterogeneously in water to yield glucaric acid, which is reduced in a H_2 atmosphere with HBr as the promoter to yield adipic acid in greater than 95% yield.[54]

Chemocatalysis of Glucose to Bioadipic Acid. As Figure 2 shows, both the glucaric acid and the FDCA routes require sequential stages of oxidation and reduction reactions. This section explains the general challenges for redox reactions during the production of bioadipic acid.

The Glucaric Acid Route. There are many studies on the use of supported precious metal catalysts, such as Pt, Au, Ru and Pd, to oxidize glucose.[12,55−57] Oxidation reactions can yield a range of products such as glycolic acid (C_2), glyceric acid (C_3), lactic acid (C_3), pyruvic acid (C_3), xylonic acid (C_5), and gluconic acid (C_6), as glucose is susceptible to undergo retro-aldol condensation to yield shorter-chained products. Only a handful of studies examined the heterogeneous oxidation of glucose to dicarboxylic acids, such as glucaric acid, xylaric acid, and tartronic acid. Hydrogenolysis of glucaric acid in the presence of metal catalysts and HI/HBr as promoters can produce adipic acid. This step requires selective activation of four internal −C−OH groups for efficient hydrogenolysis reactions.

The FDCA Route. In this route, glucose is first converted to HMF, which can be oxidized to FDCA. Compared to glucose, the oxidation of HMF or monosaccharides to FDCA was investigated extensively in the past few decades.[56,58] While Pt- and Au-based noble metals are still the preferred catalysts in many of these studies, several nonprecious metals, such as Co and Ni, also emerge as promising candidates to oxidize HMF to FDCA.[59,60] The improvement of the catalytic performance of nonprecious metals should be the focus of future studies. Notably, Dumesic and co-workers recently reported the high yield (>93%) conversion of fructose to FDCA using a Pt/C catalyst.[61] When formed, FDCA must be hydrogenated to tetrahydrofuran-2,5-dicarboxylic acid before it uses ring opening to yield a linear chain 2-hydroxyhexanedioic acid (2-HHA). Then, 2-HHA can undergo selective hydrogenolysis to become adipic acid.

Scope of the Perspective. Considering the challenges and opportunities involved in converting glucose to glucaric acid and FDCA, then to bioadipic acid, in this perspective, key advances on catalyst design, particularly on nanostructured solid catalytic materials, are first revised and summarized. The influences of catalyst composition and electronic configuration on the performances of oxidation (C−H bond activation) and hydrogenolysis (C−O bond activation) are discussed critically with representative literature results on synthesis of glucaric acid, FDCA, and adipic acid. To be more specific, this perspective is focused on the following aspects: (1) reaction mechanism: identifying key reactions with guidance on catalyst design, (2) catalyst design/synthesis: techniques and strategies for structural control, and (3) surface characterization: tailored metal−metal (oxide) interface.

Each of aspects above are discussed in two case studies: Glucaric acid route and FDCA route. Specifically, the fundamentals of oxidation and hydrogenolysis, catalyst type, performance, and design, as well as major technological challenges facing current catalytic processes, are critically discussed in detail, with the aim to provide insights into rational design of effective heterogeneous catalysts for atomic economical conversion of sugars to adipic acid from bioderived resources.

Nanocatalyst Design for C−H and C−O Bond Activation. Two major issues are known to limit the performances of the catalysts during the oxidation step in the bioadipic acid synthesis: (a) poor intrinsic activity and selectivity for dicarboxylic acids production and (b) the coproduction of smaller carboxylic acids due to the retro-aldol condensation of glucose.

With the rapid progress of nanoscience and nanotechnology, experimental studies have shown that bimetallic nanostructured catalytic materials, with tailored morphologies, tunable metal−metal interactions, and strong metal−support interactions, display promising performances to resolve both issues.

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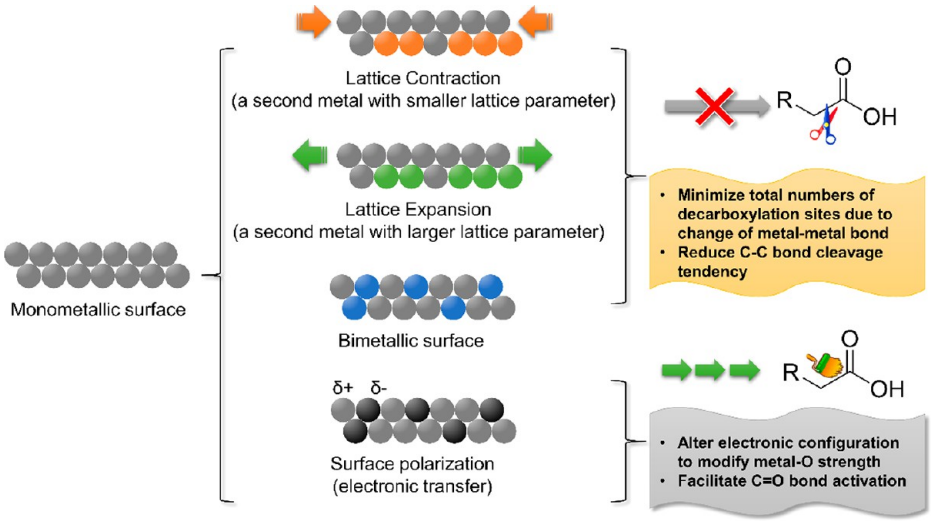


Figure 3. Conceptual design of bimetallic catalysts for enhanced performances.

Metal−Metal Interactions. One of the major challenges for oxidation of glucose and HMF is the strong tendency of decarboxylation of monocarboxylic acids (e.g., gluconic acid, furancarboxylic acid) on noble metals (C−C cleavage). In addition, intrinsically poor kinetics of the secondary oxidation reaction (C−H activation on the other end of carbon chain) poses major challenges for deep oxidation reactions.

The introduction of a second functionality (metal species) can alter the structure morphology and metal−O/metal−C bond strength due to electronic reconfiguration.[62] There are four general categories of metal−metal interactions for enhanced performances for C−H and C−O cleavage over C−C bond activation (Figure 3). The introduction of a second metal with different lattice parameters will induce lattice contraction or expansion on existing surfaces.[63−66] Some metal combinations with similar lattice distances will easily form alloy structures (e.g., random alloy or layer-by-layer morphology).[56,58] In another scenario, surface polarization caused by distinct electron affinity between two metal components lead to modified metal−O strength. Such a phenomenon can be exploited for C═O bond activation.

To achieve tunable interaction between metal sites and carboxylic groups, electronic reconfiguration is critical because electron donors induced by the presence of a second metal often leads to shifted *d* band centers for active sites. For most noble metals, interaction between active centers with carboxylic groups is weak, while the C−C bond activation is significant. This is the reason that the decarboxylation reaction can easily occur on Pt and Au surfaces. The presence of non-noble candidates such as Cu can often enhance oxidation over C−C cleavage reactions.[67] This is because those metals have strong interactions with the O part; thus, activation of the C−H bond is favorable under such conditions.

Metal−Support Interactions. Metal and support interactions have been well demonstrated in various catalytic applications such as CO oxidation, deNO_x, olefin metathesis, and semihydrogenation of acetylene.[68−74] Strong metal and support interactions are crucial for enhanced catalytic performances because the electronic coupling of metallic species with Lewis acid/base sites modifies the nature of the active sites. For example, strong interactions between Ni and acidic centers in the Al_2O_3/CeO_2 catalyst often induces synergistic enhancement in reforming of alcohols.[75−79] Strong Pd−metal interaction leads to enhanced selectivity toward olefins during acetylene semihydrogenation.[80−83]

As revealed by the literature survey, one of the key challenges in the conversion of glucose to dicarboxylic acids is the poor oxidation kinetics of the hydroxyl group in monosaccharic acids.[84] To address this issue, metal oxides with O vacancies tend to interact strongly with metal nanoparticles, thus enhancing metal−O interactions during activation of hydroxyl groups. On one hand, such unique hybridized composites display strong electron transfer from support to metallic centers. On the other hand, such a strong electrostatic force allows polarization of chemical bonds at such interfaces (Figure 4).

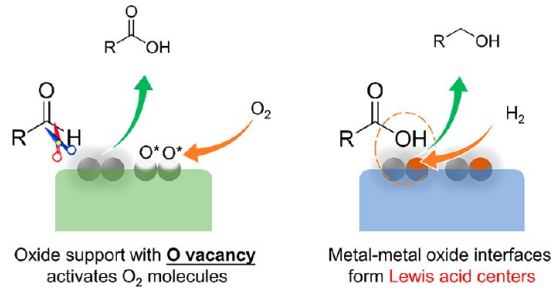


Figure 4. Conceptual design of metal−Lewis acid/base catalysts for enhanced performances.

This is because the O vacancy alters the local stoichiometric composition in metal oxides. Such evolution could create electronic enrichment over neighboring sites. The distorted local structures serve as O pools for activation of molecular O_2, while neighboring metallic sites could be coordinated through an O spillover for active oxidation reaction.

Route A: The Glucaric Acid Route. *Synthesis of Glucaric Acid: Key Steps for Oxidation Reactions.* In this route, glucose is oxidized at both ends to form glucaric acid.[85] The resulting glucaric acid then undergoes hydrodeoxygenation to form adipic acid in a reductive environment.[5,57,86]

The oxidation of glucose to gluconic acid, the key intermediate prior to glucaric acid, is a well-studied reaction, due to the various applications of gluconic acid as polyester precursors and food and pharmaceutical additives and the

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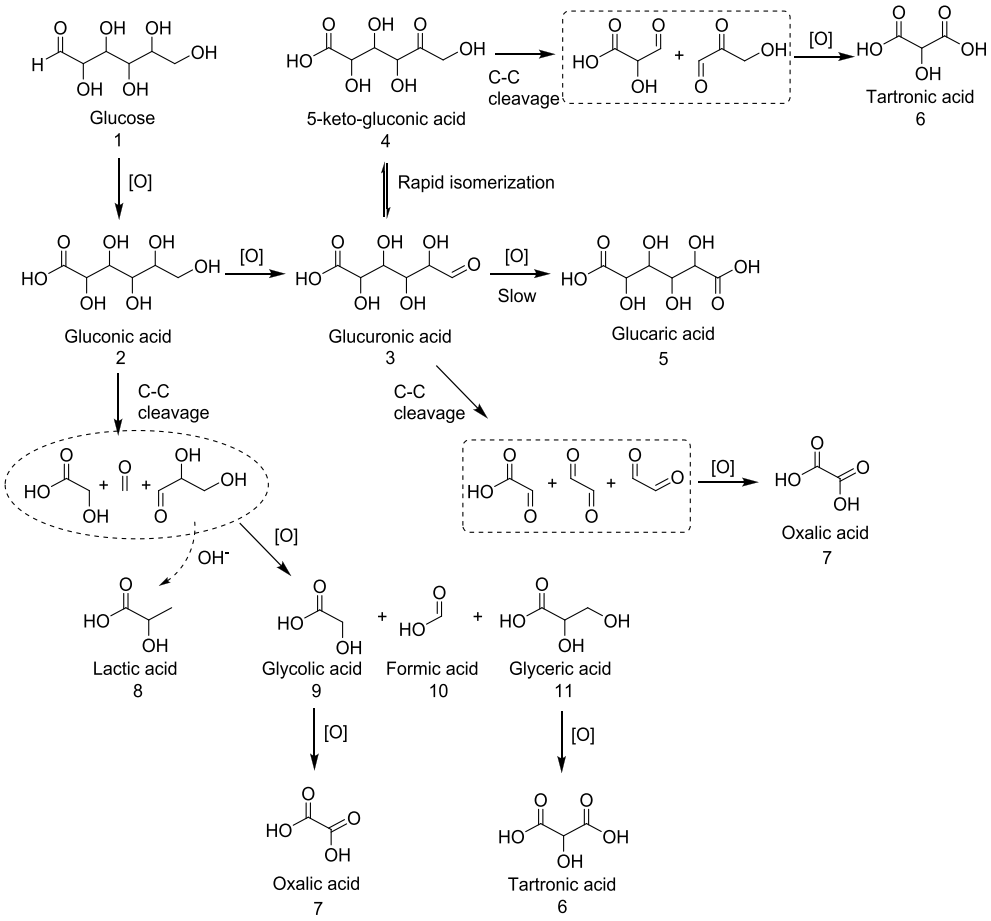


Figure 5. Reaction pathways of glucose oxidation (permission from ref 91).

Table 1. Performances of Selected Mono- and Bimetallic Catalysts for Glucose Oxidation to Glucaric Acid from the Literature

No.	Catalyst	Reaction condition	TOF (h^{-1})	X (%)	S (%)	Feature	ref
1	$PtCu/TiO_2$	60 °C, 0.1 MPa O_2; gluconic acid sodium salt	3543	100 72	25 32	Alloy structure displays high TOF values	99
2	Pt/C	60 °C, 1.2 MPa O_2; no NaOH	–	100 43	66 73	C−C cleavage very sensitive to temperature	67
3	$PtPd/TiO_2$	45 °C, 0.1 MPa O_2; continuous NaOH addition	<2000	100 41	44.1 41	Glucose inhibition on catalyst surface	91
4	$AuPt/ZrO_2$	100 °C, 4.0 MPa O_2; no NaOH	–	100 40	54 26	Au content affects glucaric acid selectivity; no characterization	98

potential development of an enzyme-free blood glucose monitor.[57] Upon oxidizing its terminal alcohol to carboxylic acids, gluconic acid becomes glucaric acid, which is also an important industrial commodity due to its biological compatibility.[87−90] It is also important to note that glucaric acid itself can be a potential precursor to replace adipic acid for nylon synthesis due to its highly hydroxyl-rich structure.[57]

To our best knowledge, only Pt and Au are known to catalyze selectively the oxidation of glucose to gluconic acid under mild conditions ($T < 100$ °C) using either O_2 or H_2O_2 as the oxidant. However, H_2O_2 is known to decompose quickly in the presence of several metal catalysts and thus is not as widely used for glucose oxidation.

On the other hand, a direct oxidation of glucose to glucaric acid is significantly more challenging as it requires transforming the hydroxyl group at the other end of glucose molecule. This is because the overall oxidation reaction rate of gluconic acid to glucaric acid is kinetically sluggish even in the presence of various precious metal catalysts, which is also accompanied by strong C−C cleavage reactions[92,93] The second step going from gluconic acid to glucaric acid is more challenging due to the stability of glucuronic acid (Figure 5). When it forms, glucuronic acid is vulnerable to rapid isomerization to become 5-keto-gluconic acid due to the presence of an aldehyde group.[91,94] Considering the competing reactions between oxidation and isomerization, it is important to alter the surface chemistry of solid catalysts with strong O-binding strength. Therefore, the

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bimetallic formulation with tunable metal−O bonding should be the preferable choice over others. Early studies also showed that isomerization, involving 1,2-hydride transfer, is a key step on acidic catalysts. Therefore, one should pay more attention on tunable electronic configuration at metal−metal interfaces with the aim to enhance intrinsic activity for secondary oxidation of glucose.

Another pitfall is that glucuronic acid may undergo C−C cleavage reactions via either a retro-aldolization to yield C_2−C_4 products in the presence of base or acid promoters or may decarboxylate to yield CO_2 over Pt−Pt and Au−Au sites.[95−97] Enhancing oxidation of aldehyde groups in glucuronic acid to glucaric acid, rather than the C−C cleavage reaction, is very challenging, particular in basic/acidic conditions, where retro-aldolization is difficult to restrain (Table 1).[93,98] Therefore, rational design of an effective catalyst should be primarily focused on enhancing the interaction of C═O groups on metallic surfaces against desorbing to aqueous media, where retro-aldolization occurs much more easily. Glucose, gluconic acid, and glucaric acid have been observed to undergo C−C cleavage to yield small hydroxyl acids, such as formic acid, glycolic acid, glyoxylic acid, glyceric acid, lactic acid, and pyruvic acid, according to the experimental results from our recent studies.[91,99] These fractionated organic acids are commonly observed as coproducts during glucose oxidation in the presence of acidic/alkaline media. This is because basic conditions may enhance C−C cleavage via retro-aldolization, while acidic media may enhance decarboxylation reactions catalyzed by metallic sites. Therefore, it is clear that maintaining nearly neutral pH could possibly reduce side reactions leading to C−C cleavage and thereby increase the yield of glucaric acid.

Besides glucose, selective oxidation of pentose, such as xylose to xylaric acid, a dicarboxylic acid, is also of great chemical interest.[100,101] Despite the difference of one carbon unit, it is strongly believed that the oxidation of C_5 sugar to C_5 dicarboxylic acid follows similar reaction pathways and therefore experiences similar challenges.

Bimetallic Catalysts for Enhanced Oxidation. *Pt-Based Catalysts.* Oxidation of glucose to glucaric acid in aqueous media has received rising attention since the 1980s.[102,103] Dirkx and co-workers found that Pt/C catalysts can facilitate oxidation of glucose to gluconic acid and glucaric acid in alkaline media. Experimental studies confirmed that primary (to gluconic acid) and secondary (to glucaric acid) oxidations of glucose are both catalyst structure-dependent reactions. For example, our recent work showed that Pt-based bimetallic catalysts (see Figure 6(a, b, c) for preparation details), including PtCu, PtCo, and PtPd, show good performances in oxidation of gluconic acid (in the form of sodium salt) to glucaric acid, tartronic acid, and other carboxylic acids in the presence of base promoters.[99] In particular, $PtCu/TiO_2$ catalysts with alloy and core−shell morphologies display different activity and selectivity toward glucaric acid (entry #1, Table 1). The alloy $PtCu/TiO_2$ catalyst takes 3542.9 h^{-1} for conversion of sodium gluconate at 60 °C in the presence of a NaOH promoter, while core−shell structures exhibit much lower activity for oxidation reactions. However, the bimetallic $PtCu/TiO_2$ catalyst did not show good selectivity for direct glucose oxidation to glucaric acid (~25%). According to surface characterization, the addition of Cu could enhanced the C−OH oxidation to C═O, but it also facilitated C−C cleavage *via* retro-aldolization under reaction conditions. Therefore, byproducts of C_2 and C_3 mono-alphahydroxy acids from direct decomposition of glucose were formed in

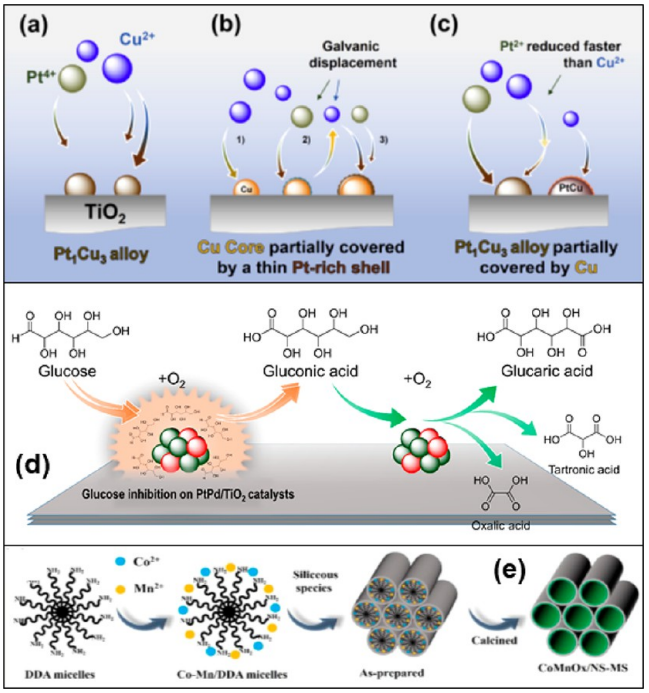


Figure 6. Preparation of catalysts for oxidation reaction of glucose. (a, b, c) Preparation of $PtCu/TiO_2$ catalysts (permission from ref 99), (d) substrate inhibition of glucose on $PtPd/TiO_2$ catalyst (permission from ref 91), (e) preparation of mesoporous Co−Mn oxide catalysts (permission from ref 107).

abundance. To minimize the unwanted decomposition of glucose, Shi and colleagues later observed that the yield for glucaric acid can be enhanced to approximately 67% in the absence of base promoters with the same bimetallic $PtCu/TiO_2$ catalysts, suggesting that an acidic environment can be effective to suppress retro-aldol C−C cleavage (entry #2, Table 1).[67]

It was later found on bimetallic $PtPd/TiO_2$ catalysts that glucose undergoes substrate inhibition during oxidation processes (entry #3, Table 1; Figure 6(d)).[91] The presence of glucose prevents secondary oxidation; thus, gluconic acid is the only product observed at the beginning. Once glucose is completely converted, glucaric acid and products with shorter carbon chains generated from C−C cleavage will occur in the reaction media. It is highly possible that the presence of Pd enhances interactions between a metal catalyst and the −C═O group, leading to significant substrate inhibition under the reaction conditions. The presence of Pd in the Pt−Pt lattice actually minimizes the total amounts of active sites for strong Pt−C═O interactions. More importantly, Pt catalyzed C−C cleavage through decarboxylation can also be minimized.[104−106] It is believed that the substrate inhibition effect significantly reduces decomposition of glucose; thus, glucaric acid yield is much higher (yield ~44%), although the overall activity is not as good as the $PtCu/TiO_2$ catalyst. Conversion selectivity studies also showed that glucaric acid seems not to decompose under such conditions (entry #3, Table 1) because selectivity has not been lowered with enhanced conversion of glucose. In the meantime, how the Pt−Pd interaction leads to substrate inhibition is still unclear at this stage.

Vlachos and co-workers studied the activity and selectivity of Pt/C catalysts for glucose oxidation in base-free media. They observed that glucaric acid yield is 74% (TOF: 879 h^{-1}) at 80 °C and 1.38 MPa O_2.[94] Different from $PtPd/TiO_2$ catalysts, there is

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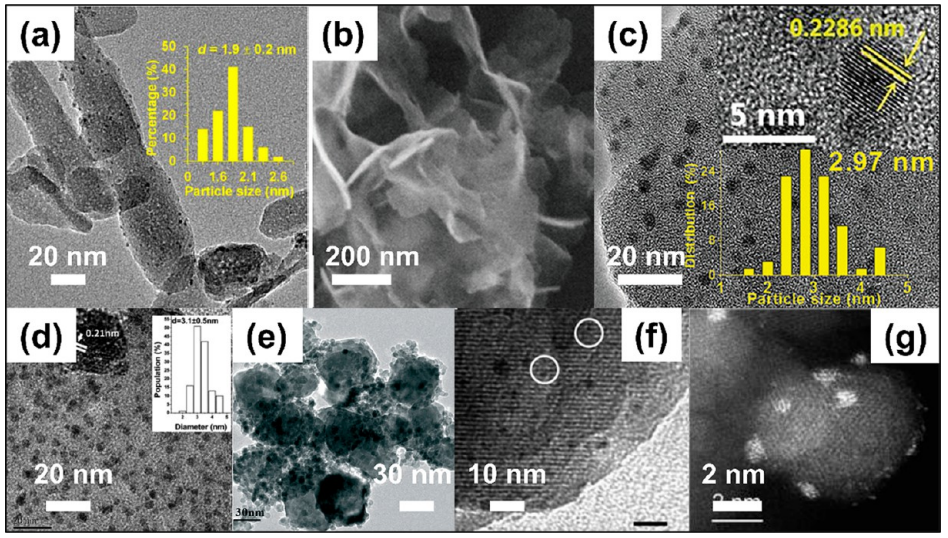


Figure 7. Structural characterization of (a) Ru/HAP catalysts (permission from ref 112), (b) nanoflake PdNi/$Mg(OH)_2$ (permission from ref 113), (c) AuPd alloy on carbon (permission from ref 114), (d) Pd/hydrotalcite catalyst (permission from ref 115), (e) $Fe_xZr_yO_z$ catalyst (permission from ref 116), (f) Au/NaY catalyst (permission from ref 117), and (g) Au/FeO_x surface (permission from ref 118).

no substrate inhibition effect of the glucose molecule on the Pt/C surface. They found that gluconic acid is the precursor for glycolic and glyceric acids, while the C−C cleavage of glucaric acid leads to formation of oxalic and tartronic acids, which is confirmed by decreased selectivity of glucaric acid with increasing conversion of glucose. It was also found that increasing the initial pH from 7.2 to 13.2 induces C−C cleavage reactions with selectivity toward tartronic and oxalic acids from 0% to 26%, while glucaric acid is very low (9%) with an initial pH of 13.2.

It is interesting to find that the activation of molecular O_2 has not been fully discussed with activation of C−OH and C═O bonds in this area. But various work has provided evidence for metal−support interactions. It is worth noting that the defective surfaces of metal oxide supports may play a key role in activation of molecular O_2 at metal−support interfaces. Thus, oxidation rates can be significantly enhanced.[108] As hot electrons may be transferred from cationic (support) to metallic sites, they eventually migrate to O_2 molecules for oxidation reaction. Clearly, such a step is critical for improved oxidation activity. In terms of selectivity toward activation of C−OH and C═O rather than C−C cleavage, metallic sites should be more crucial, according to the catalyst characterization shown in the current literature.

From a process point of view, side products generated from hydrolysis of cellulose and hemicellulose might be detrimental for catalyst stability. For example, furan compounds are often formed during further transformation of sugars. This issue needs to be addressed for successful implementation of the conversion process of sugars. The impurities from hydrolysis of cellulose as well as the side products generated during glucose oxidation need to be removed for durable catalytic systems.[98,109,110] Enzyme residuals in crude glucose often induces isomerization of sugars and poisons metal catalysts, leading to poor selectivity toward oxidation products. Besson and co-workers studied the influence of residues in softwood hemicellulose hydrolysates on catalytic activity, selectivity, and stability of Pt/C catalysts during the oxidation process.[111] They confirmed that residues such as sodium salts and unconverted disaccharides will not inhibit catalyst activity, while the presence of furan aldehydes significantly slow reaction rates. Small phenolic residues from lignin showed a detrimental effect and should be eliminated prior to catalytic oxidation of glucose.

Au-Based Catalysts. To our best knowledge, the overall performances of Au-based catalysts are not as good as Pt ones, although various studies also confirm the structure sensitivity of Au nanoparticles on oxidation activity (characterization in Figures 7(f, g)). The addition of Bi and Cs as promoters could enhance the oxidation activity of Au catalysts.[93,119] However, the C−C cleavage reaction of glucose and gluconic acid cannot be ignored over Au catalysts. Carbon-supported Au nanoparticles 2 nm in size display relatively higher oxidation activity (TOF: 480 h^{-1}) for glucose oxidation compared with ones less than 11 and 5−9 nm in size.[93] Selectivity toward glucaric acid is low (S ∼ 31%) on these Au/C catalysts at 60 °C and 1 MPa O_2. Different from Pt-based catalysts, a higher NaOH:glucose molar ratio (3:1) is necessary for complete conversion and improved selectivity to glucaric acid over Au catalysts.

Combining Au with Pt catalysts displays surprisingly remarkable performances for oxidation of C_5 and C_6 sugars. Although detailed surface morphologies and electronic configurations are not illustrated in their work, Besson and co-workers confirm the promising performances of AuPt/ZrO_2 with several-fold enhancement compared with Pt/C in oxidation activity of aldoses (entry #4, Table 1).[98,109] Besson and co-workers' work also compared the catalytic performances of various bimetallic catalysts including AuCu/TiO_2, AuAg/C, AuPd/C, and AuPt/C.[111] But the Pt/C catalyst is still the best one among others in terms of glucaric acid yield (54%).[98]

The pH of the reaction medium is another critical criterion in controlling the selectivity of glucaric acid. In general, higher pH (∼10) enhances the oxidation rate on Pt and Au catalysts, but at the same time, it has been observed that non-neutral pH may promote C−C cleavage. On the basis of our literature survey, numerous previous work has already shown that the presence of alkaline promoters such as NaOH and KOH can substantially enhance the activity of noble metal catalysts, owing to the electronic reconfiguration of metallic centers in the presence of OH^- ions. But for glucose oxidation, strong alkaline conditions undesirably enhance decomposition reactions. Therefore, finely

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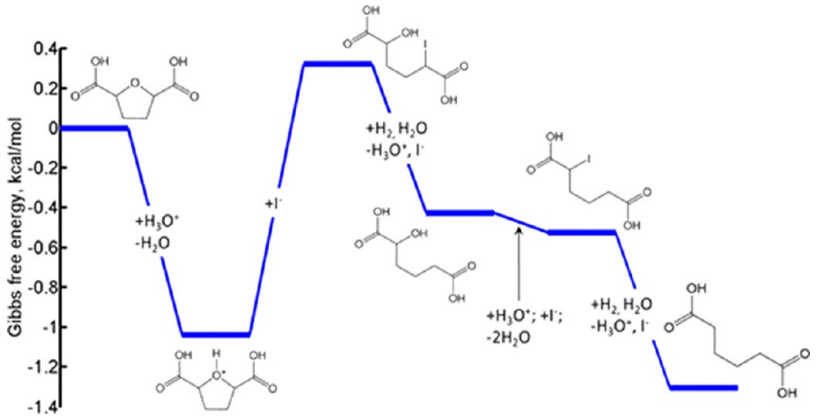


Figure 8. Detailed reaction pathway for conversion of FDCA derivative to adipic acid (permission from ref 131).

tuned pH in acidic and weak basic conditions is important in this area. Recent work by Repo and co-workers show that the oxidation of glucuronic acid selectively gives glucaric acid (S > 99%) in the presence of a Au/Al_2O_3 catalyst.[120] Increasing the pH of the reaction medium only increases the oxidation rate, while no byproduct is formed during the reaction.

Conversion of Glucaric Acid to Adipic Acid. The literature on hydrodeoxygenation of bio-oxygenates has demonstrated that terminal and internal −OH groups could be selectively removed in the presence of metal catalysts. Initial attempts on direct hydrogenolysis of glucaric acid has achieved limited success, as the C−O cleavage reaction hardly occurs without additives even under harsh conditions (T > 220 °C).[5] It is found that the addition of hydrogen halide promoters (HBr, HI) could facilitate this reaction by rupturing the C−O bond and forming the R−X intermediates, followed by the generation of adipic acid in the aqueous phase. Interestingly, it seems to be the only effective way to generate adipic acid from glucaric acid so far.[121]

Toste and co-workers developed a deoxydehydration route for hydrogenolysis of glucaric acid to adipic acid.[122] They found that Pd/C catalysts selectively remove −OH groups in glucaric acid in the presence of $KReO_4$ as the homogeneous catalytic promoter. Glucaric acid tends to undergo internal dehydration catalyzed by ReO_4^-, while hydrogenation of the C═C bond over Pd/C catalysts can be realized at 150 °C with low H_2 pressure (<0.5 MPa). The influences of catalyst composition, morphologies, and metal−support interactions are still unexplored in this area.

Comments and Perspectives. Most of the existing studies on oxidation and hydrogenolysis catalysts on the glucaric acid route are primarily focused on performance tests of mono- and bimetallic catalysts. While only a few studies on bimetallic nanocatalysts reveal the crucial role of metal−metal interactions on enhanced kinetics for C−O and C═O bond activation, very limited results have been published for the fundamental understanding of metal−metal oxide interfacial electronic configurations on oxidation and hydrogenolysis steps.

For the oxidation step, Pt- and Au-based materials are promising candidates being studied to catalyze glucose conversion to glucaric acid. Other candidates such as Co and Cu also show good selectivity but poor intrinsic activity; thus, they are often employed as promoters for noble metals. Bimetallic PtCu, PtPd, and AuPt catalysts with different compositions show better catalytic activity and selectivity compared with monometallic Pt and Au materials. However, how size, shape, and metal−support interactions affect the intrinsic activity of metal catalysts is far from fully understood.[55,123] In addition, despite exciting results on developing alternative candidates for effective oxidation of glucose to glucaric acid, the kinetics have yet to be truly studied by researchers.[46,124−126] This is because most attention has been paid to understanding how catalyst composition affects the oxidation of glucose, also with limited knowledge on the significance of each pathway. Bridging the gap between well-characterized bimetallic or metal−acid-hybridized catalysts, with fundamental understanding on activation of C−H and O−H bonds, is needed in this area. There is an urgent demand for comprehensive kinetic modeling to assess the contributions for each reaction involved in the complicated network.

As discussed above, there is a lack of systematic investigation on the hydrogenolysis mechanism. Although existing patents reported promising results, the reaction scale is too small, and it is very difficult to reproduce their experimental results. There is no doubt that more research efforts should be paid on manufacturing effective hydrogenolysis catalysts for selective removal of −OH groups in glucaric acid.

Route B: FDCA−Adipic Acid Route. *General Information on Reaction Pathways.* Glucose can also be transformed into hydroxyl methyl furfural (HMF) through isomerization and dehydration, which is then oxidized to yield FDCA. Compared to the glucaric acid route, the FDCA route carries a greater research interest, as the chemical potentials extend beyond adipic acid preparation. For instance, FDCA is also an emerging building block to replace petroleum-derived *para*-terephthalic acid for polyester plastics production.[127,128] Upon hydrogenation of the aromatic furan ring, FDCA can undergo a series of hydrogenation reactions to yield adipic acid Figure 8.[129,130]

Compared with the studies on glucose oxidation to glucaric acid, there are tremendous efforts for HMF conversion to FDCA.[4,5,8,12,56,57,86,128,132−134] FDCA was also identified as one of the top 12 platform chemicals derivable from biomass in a 2004 report by the U.S. Department of Energy.[86,128,135] This receives wide attention as it can be transformed into various polymer materials, such as polyamides, polyesters, and polyurethanes,[133] to replace PET plastics for the chemical industry. Sustainable production of FDCA constitutes a substantial step forward in terms of renewable products, which opens a new bioderived chemical market. It is important to note that adipic acid was not regarded as a top platform chemical from biomass. However, from the molecular structure point of view, FDCA can be promising in synthesizing adipic acid as a ring-opening

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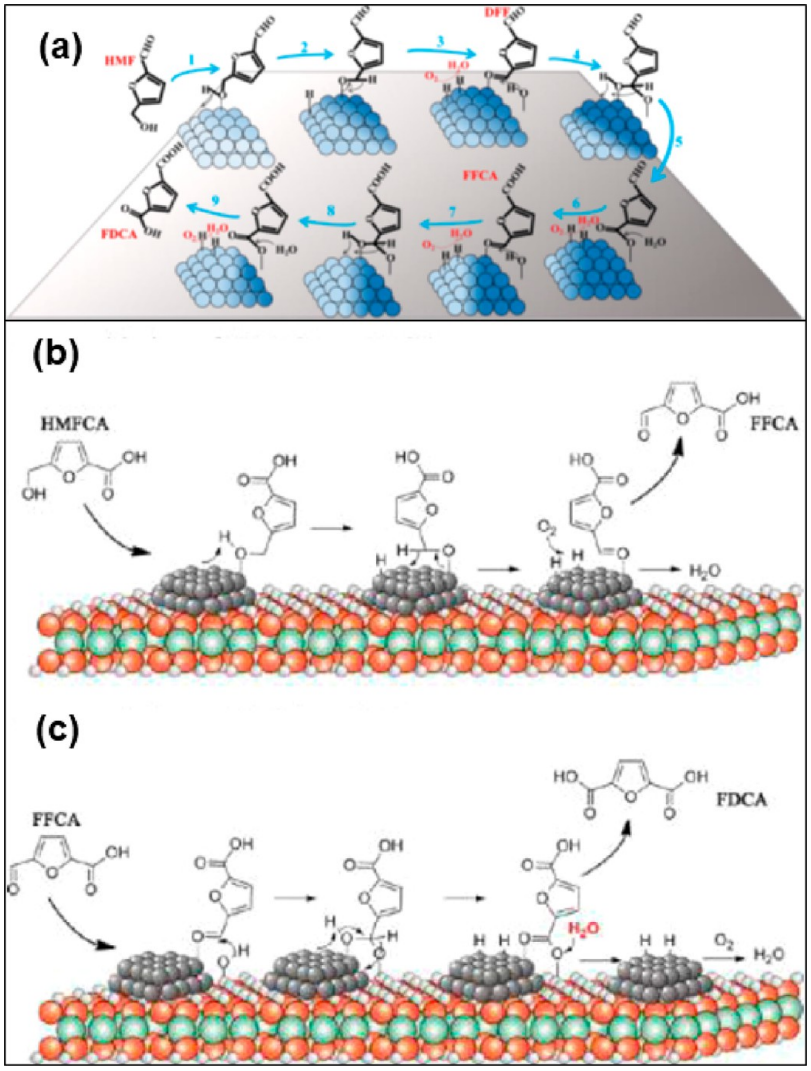


Figure 9. Possible reaction mechanism for HMF oxidation to FDCA: (a) reaction mechanism of HMF oxidation on Ru catalyst (permission from ref 112) and (b, c) HMF oxidation of FDCA over Pd/hydrotalcite catalyst (permission from ref 115).

reaction, and hydrogenation can occur under mild conditions. In view of producing biobased nylons, deoxygenation of FDCA leads to the formation of adipic acid in the aqueous phase. Therefore, this conversion route also receives increasing attention in recent years.

Conversion of glucose to 5-HMF has been a highly explored topic in recent years, and the reader may refer to other references provided for more details.[38,136−139] Earlier work on converting HMF to FDCA was also focused on using stoichiometric agents such as $KMnO_4$ or HNO_3. Recent publications on HMF oxidation described performances of Au, Pt, Pd, and Ru metals in the aqueous phase, mostly with alkalis as promoters.[56,58,86,140] In this section, we are primarily focused on how bimetallic catalysts and strong metal−support interactions enhance oxidation and hydrogenolysis reactions.

Synthesis of FDCA: Oxidation and Catalysts. Preliminary work published by Davis and co-workers has already confirmed that O in water is the primary source of O atoms during oxidation reactions on noble metal catalysts.[141] There are two routes for FDCA formation in an oxidative environment. As shown in Figure 2, activation of the C═O bond in HMF generates 5-hydroxymethyl-2-furancarboxylic acid (HMFCA) as a primary intermediate. Secondary oxidation activates the C−OH bond on the other end. Further oxidation results in the formation of 5-formyl-2-furancarboxylic acid (FFCA), and FDCA is eventually formed as the final product. Oxidation of HMF catalyzed by Pt, Pd, and Au materials is usually found to follow this route (Figure 9). Alternatively, activation of a C−O bond leads to generation of 2,5-dihydromethylfuran as an important intermediate, which is often observed on Ru, Mn, and Co catalysts.[4,5,8,12,56,57,86,128,132−134]

Metal−Support Interactions. Metal−support interactions are important for O storage with tunable redox properties.[142,143] For example, CeO_x can be used as a redox promoter to facilitate electron transfer of AuCu catalysts during HMF oxidation.[144,145] Other work also confirms the importance of surface redox sites in enhancing the FDCA yield to 95+% in alkaline media.[146,147] Besides Ce^{3+}/Ce^{4+} pairs, doping Bi^{3+} and Mn^{2+} seems to create more O vacancies; thus, the turnover rate of the oxidation reaction can be significantly enhanced.[148,149] Such a vacancy is often believed to act as the O pool for activation of C−H bonds in HMF molecules.

Electron transfer from metallic centers to oxide supports will be important for modifying electronic properties of the metal center, thus improving the efficiency in O_2 activation.[107,144,145,150] For example, Xu and co-workers characterized

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Table 2. Key Results on HMF Oxidation to FDCA over Nanostructured Catalysts[a]

No.	Catalyst	Reaction condition	TOF (h^{-1})	X (%)	S (%)	Feature	ref
1	Au/HY	60 °C, 0.3 MPa	15.8	100	>99	Size: 1 nm, supercage confinement	117
2	AuPd/AER	100 °C, 1.0 MPa	–	100	98	AuPd−N interaction	155
3	AuPd/LaCaMgAl−LDH	100 °C, 0.5 MPa	8.3	96.1	89.4	Strong basicity of LDH when doped with Ca and La; formation of Au^{n+}	157
4	Ru/HAP	140 °C, 1.0 MPa	4.0	100 80	99.9 25	Acidity-basicity tuned by varying Ca/P ratio	112
5	$Ru/MnCo_2O_4$	120 °C, 2.4 MPa	33[b]	100	99.1	Strong Bronsted acidity of $MnCo_2O_4$	158
6	$PdNi/Mg(OH)_2$	100 °C, air flow	1.6	99	98	Nanoflake $Mg(OH)_2$ is stable	113
7	PdNi/C	80 °C, air flow, Na_2CO_3	–	>99	86	PdNi alloy	159
8	PtNi/C	100 °C, 0.4 MPa	24.3	100	~95	Pt skin on Ni surface	160
9	Pt/N-doped carbon	110 °C, 1.0 MPa	13	99 62	94 33	Pt−N interaction	156
10	CeO_2−MnO_2	110 °C, $KHCO_3$	–	>99	~90	O transfer to lattice activates MnO_2 sites	148

[a]AER, anion exchange resin; HAP, hydroxyapatite; LDH, layer doubled hydroxide. [b]TOF unit: mmol/g-Ru/h.

surface acidity of HY-, NaY-, and NaY-confined Au nanoclusters for HMF oxidation (entry #1, Table 2; Figure 7(f)).[117] They found that Au clusters interact strongly with surface and internal (inside supercage) −OH groups; thus, −O−Au sites can be formed. Therefore, it is clear that selectivity toward FDCA (99%, at 60 °C) can be well tuned by altering the Au interaction with the support. The particle size of Au was not well correlated with the FDCA yield; thus, the cationic effect might be significant in their work. Experimental results also show that the presence of a mesoporous CeO_x framework also contributes to improved oxidation performances.[149,150] Similarly, Fraga and co-workers observed that the Pt−$Mg(OH)_2$ interaction is critical for enhanced oxidation performances in slightly basic media. But the Mg^{2+}−H_2O interaction displays a detrimental effect on catalyst stability.[151] In parallel studies, various researchers proposed that morphologies of $Mg(OH)_2$ might also play an important role in stabilizing metallic species as weak basic sites that are durable during FDCA formation.[59,113,140] Another example involving experimental studies on the interface of AuPd/$Zn(OH)_x(CO_3)_y$ implies that the unique structures of support can be stabilized in aqueous media and prevents metal sintering.[152] In the above-mentioned studies, the contribution of structural defects and pore confinement needs to be further confirmed, as enhanced FDCA yield cannot be simply ascribed to changes of one of these parameters.

Some additional efforts have been focused on using ligands to manipulate interactions at Au−metal oxide interfaces.[144] Satsuma and co-workers discussed the influence of FeO_x modification for Al_2O_3 with atomically dispersed Au species.[118] The high point of zero charge for FeO_x species interacts strongly with $Au(OH)_3Cl^-$ species; thus, Au dispersion can be enhanced due to lattice strain at the interface. Using a similar methodology, Au^{n+}−Ce^{3+}/Ce^{4+} interactions can be manipulated by altering the mesoporosity of CeO_x structures. The defects at the Ce^{3+}/Ce^{4+} interfaces tend to bind AuO_x species; thus, both dispersion and activity of Au catalysts can be improved.[150] Another example reported by Nakajima and co-workers shows that Ce^{3+} tends to stabilize −OH groups in HMF molecules; thus, side reactions to form humin can be reduced to a large extent.[153] Obviously, −OH groups in hydroxide supports tend to interact much stronger with metal species. For example, Ni and Fe species in layered double oxides often exist in cationic or hydroxide form and can barely be reduced to the metallic state.[154] The compact coordination with the hydroxide framework alters the electron-withdrawing capability. Such unique structures are often found to be crucial for enhanced FDCA selectivity (S: 98%) during oxidation of HMF.

Besides the O vacancy strategy, the electrostatic interaction of metal nanoparticles with amine groups on the catalyst surface also shows promising features for robust performances in HMF oxidation. This is because the extra electron pairs often stabilize nanoparticles via a strong electronic static force on the catalyst surface. Cho and colleagues developed an ion exchange resin-supported bimetallic AuPd catalyst with excellent yield for FDCA.[155] Although catalytic activity was not compared with existing results, the aromatic ring−Pd^{2+}/Au^{3+} interaction induces entrapment of Au and Pd species, anchored by amine groups in the framework, leading to almost a 98% yield of FDCA at 100 °C and 1.0 MPa O_2. Similar results reported by Wang and co-workers confirmed that strong Pt−N interactions in N-doped carbon could increase the overall activity for Pt catalysts (entry #11, Table 2). The details on pyridine-, pyrrole-, and graphite-type N and their interactions with Pt nanoparticles are yet to be further explored.[156]

Compared with other noble metals, Ru is less effective to carry out oxidation reactions. However, by tuning metal−support interactions, good yield of FDCA can be obtained. Fang and co-workers observed that by rationally controlling the Ca/P ratio in hydroxyapatite, conversion of HMF to FDCA on Ru catalysts can be improved from less than 34% to 99.9%.[112] Kim and colleagues developed spinel-structured $MnCo_2O_4$ materials as solid supports for immobilizing Ru nanoparticles.[158] They found

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that the surface Bronsted acidity of $MnCo_2O_4$ materials has a strong adsorption affinity to O. The yield of FDCA is 99.1% (TOF: 33 mmol/g-Ru/h at 120 °C).

Besides noble metals, some recent work also paid attention to non-noble-based solid catalysts.[58,161−163] For example, Yadav's group compared performances of various spinel-structured mixed metal oxides, such as $CuFe_2O_4$, $CoFe_2O_4$, $MnFe_2O_4$, and $MgFe_2O_4$, for FDCA synthesis from HMF.[164] Although the yield of FDCA (~90%) is much lower than Pt- and Au-based catalysts, these catalysts exhibit promising features for C═O and C−O bond activation in the presence of peroxides as oxidants. Novel structured Fe_2O_3@Mo(Pd)−hydroxyapatite[165,166] and Fe_3O_4−CoO_x catalysts[167] were also found to be very effective in FDCA synthesis (HMF conversion, 100%; FDCA selectivity, 15%−93%). Wang and colleagues correlated the surface concentration of Mn^{4+}/Mn^{3+} and Ce^{4+}/Ce^{3+} in mixed CeO_2−MnO_2 oxide catalysts with oxidation rates of HMF and FDCA selectivity (entry #10, Table 2).[148] They found that strong Mn^{4+} and Ce^{4+} interactions induced formation of lattice oxygen O_2^{2-} and defect oxygen O^- in oxide structures with synergistic oxidation rates and enhanced FDCA yields (~90% at 110 °C), while harsh calcination temperatures above 500 °C leads to phase segregation with poor oxidation performances. Unfortunately, most existing studies are primarily focused on surface properties including the size and shape of metal particles, as well as surface facets of heterogeneous supports, rather than cation−support interaction mixed metal oxide catalysts.[168]

Metal−Metal Interaction. Particle size and morphology seem to be less critical factors for conversion of HMF to FDCA.[144] Davis and co-workers have summarized recent studies on the possible size effect of Au, Pt, and Pd nanoparticles on catalytic activity for HMF oxidation.[169] They found that supported Au on C, TiO_2, Fe_2O_3, and CeO_2 display completely different TOF values (360−1100 h^{-1} at 130 °C), although particle size is very similar on these supports. Experimental studies on the influence of size is also complicated by introducing stabilizers, as some of the surface active sites on catalyst surfaces are believed to be blocked. Classic approaches such as ligand-mediated growth of nanoparticles are still popular for tunable selectivity for FDCA formation.[170] Therefore, the compositional effect on catalytic activity and selectivity of bimetallic catalysts has also been studied in recent work. Classic oxidation catalysts such as Au and Pt are still the primary focus for HMF oxidation. Different from the case of the glucaric acid route, where decarboxylation is significant over noble metals, in this part, the addition of a second metal often alters the electronic configuration; thus, the intrinsic oxidation activity for −C−OH and −CHO activation can be enhanced.[171] For example, Fu and colleagues developed an atomic deposition method to coat Pt skin on Ni nanoparticles with almost 8-fold enhancement for oxidation rates of HMF.[160] Experimental work also confirmed that Au and Pd metals catalyze HMF oxidation to FDCA through different reaction pathways due to intrinsically different activation mechanisms of C−OH and C═O at metal centers.[171,172] Batch experimental studies confirm that while oxidation of HMF over Pd catalysts only gives HMFCA as main products, the introduction of Au-catalyzed secondary oxidation forms FDCA.[173] Surface characterization of bimetallic AuPd systems indicates that Au lattices tend to cover Pd surfaces; thus, synergistic oxidation activity can be observed. Alloying Au and Pd with other metals also shows promising results for enhancing the HMF oxidation rate. For example, PdNi (10:90 atomic ratio) catalysts display almost 5-fold enhancement for the HMF oxidation rate (entry #7, Table 2).[159]

With regard to the kinetics of an oxidation step for HMF to FDCA, experimental studies have confirmed the fast rates for primary oxidation to monocarboxylic acid, while formation of FDCA is still the rate-determining step for deep oxidation of HMF. The limited experimental studies have also shown that oxidation of HMF is initiated with activation of −OH groups rather than −C═O groups. The overall kinetics are much slower.

Another aspect is the activation of O_2. In this part of the study, a variety of Pt, Pd, and Au catalysts display strong interaction with oxide supports due to the O vacancy available on the surface. However, most experimental work did not particularly discuss the role of an O vacancy on activation of O_2 rather than the HMF molecule. Further attention should be paid in this area, with the aim to improve overall activity.

Conversion of FDCA to Adipic Acid: Hydrogenolysis and Catalyst. Similar to hydrogenolysis of glucaric acid, limited experimental results exist in the literature for FDCA conversion to adipic acid in the presence of solid catalysts.[12] Bimetallic acidic candidates, including PtMo and RhRe, were employed to conduct mechanistic studies for adipic acid formation.[5] Vlachos and co-workers studied the influence of iodide salt on the C−O cleavage of FDCA over acidic catalysts.[129] They found that the C−O cleavage of FDCA does not require metallic catalysts even under mild conditions, as the attack of the O site in the furan ring by H^+ is kinetically favorable, with ring-opening reactions catalyzed by I^- as the rate-limiting step.[131] They proposed that homogeneous HI species facilitate C−O cleavage via direct protonation at acidic sites through an ion exchange process. Kinetic modeling reveals that nucleophilic substitution forming the C−I bond might be the rate-limiting step during FDCA conversion. Replacing HI with a solid acid catalyst affects the delivering rate of protons, and thus, the C−O cleavage reaction rate would be manipulated. Again, the quantitative assessment on the reaction rate is not available in the current literature.

Comments and Perspectives. A strong metal−support interaction is considered a more promising technique to manipulate catalytic activity and selectivity for metal−metal oxide materials during HMF oxidation. An O vacancy is known to anchor active O species, while structural defects tend to bind and stabilize cationic metallic species during activation of HMF molecules. Despite extensive efforts, the nature of active centers on the impact of C−OH activation has yet to be well documented in the current literature. In addition, how well-defined bimetallic and metal−acid hybrids affect the intrinsic kinetics of HMF oxidation is far from preliminary studies. This issue might be critical for further development of HMF conversion to adipic acid technology. Although they might not be the topic in this work, the mass transfer and kinetics for oxidation of HMF need to be addressed in technological development.

With regard to pH values of reaction media, as already mentioned in a previous section, a strong alkaline medium is beneficial for enhanced HMF oxidation to FDCA. Early studies for both noble and non-noble metals had been primarily conducted in basic media. However, recent experimental work has shown that noble metal catalysts also show good performances in neutral or acidic media. But elevated pressures (>1 MPa) or basic supports are still needed.

In aqueous phase oxidation of bioderived oxygenates, nonprecious metal catalysts become increasingly popular, as

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the high production costs of FDCA and other carboxylic derivatives are known as bottlenecking issues for the sustainable production of bioderived adipic acid and derivatives.[57,58,174] Inexpensive metal oxides are not new in terms of structural defects. The nature of cation−cation interactions, particular in the presence of protic solvent molecules, demands extensive attention in academia and industry in order to develop active and durable catalytic materials. In addition, developing scalable approaches which could tailor nanostructures of mixed metal oxide materials might provide key technological breakthroughs in this area.

Other Emerging Routes. *1,6-Hexanediol as Key Intermediate: Revisit the Alternative Route.* Catalytic hydrogenation of HMF to 1,6-hexanediol (1,6-HDO) has drawn particular interests as conversion of HMF to transportation fuels is considered a hot topic in biomass conversion.[56,175,176] Despite a decade of effort, direct conversion of HMF to 1,6-HDO still remains a grand challenge in this area, as it is difficult to find effective catalytic materials for simultaneous saturation of C=C bonds while promoting ring openings of the cyclic ether.[177] 1,6-HDO is conventionally produced via hydrogenation of adipic acid. With emerging technologies for biorefineries, 1,6-HDO can be readily available as a downstream intermediate during HMF conversion.[11,56] As competing reaction pathways against other hydrogenation routes (see Figure 10 for details), 1,6-HDO

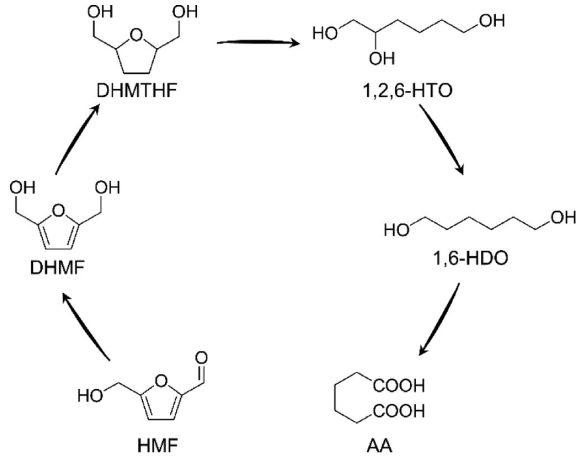


Figure 10. General reaction pathways for HMF conversion to adipic acid (DHMF: 2,5-dihydroxymethylfuran; DHMTHF: 2,5-dihydroxymethyltetrahydrofuran; 1,2,6-HTO: 1,2,6-hexanetriol; 1,6-HDO: 1,6-hexanediol).

formation needs saturation of the furan ring and a subsequent ring-opening reaction to form 1,2,6-hexanetriol as the key intermediate. Therefore, dual functional catalytic sites with both C=C cleavage and ring-opening activity are considered as preferable candidates for 1,6-HDO synthesis from biomass.

Monometallic noble catalysts display superior activity and selectivity for C−O cleavage; thus, dimethyl furan can be formed as the key product during HMF conversion.[178] Literature results have demonstrated the remarkable performances of bimetallic catalysts and metal−metal oxide hybrid materials for selective ring-opening reactions of HMF and derivatives.[58] For example, Huber and co-workers reported that Pt−WO_x hybrids display remarkable selectivity toward 1,6-HDO, as the unusual formation of Bronsted acidic sites is significant due to strong interaction between Pt and W=O sites.[179] Ring-opening reactions of 2,5-dihydroxymethyltetrahydrofuran to 1,6-HDO are facilitated by almost 10-fold at such Pt−WO_x interfaces according to the experimental results. Catalyst tests on classic $RhRe/SiO_2$ and $IrRe/SiO_2$ catalysts suggest that ring opening and saturation of the C=C bond might be competitive on bimetallic catalyst surfaces, as direct conversion of HMF to 1,6-HDO does not proceed even under harsh conditions.[180] However, the reaction rate of 2,5-dihydroxymethyltetrahydrofuran conversion to 1,6-HDO is almost 10-fold higher than that of HMF to 1,6-HDO. Therefore, HMF conversion to 1,6-HDO often requires two steps to achieve good product yield.

Inexpensive bimetallic catalysts such as NiCo also exhibit promising performances as cooperative effects between Ni and Co species enhance ring-opening reactions.[58] With Co facilitating C−O cleavage of HMF, the addition of Ni as a promoter alters the orientation of HMF adsorption; thus, ring-opening reaction rates were increased by several folds. Similar concept has also been demonstrated to the Pd^0−Pd^{n+}/zirconium phosphate (ZrP) catalyst system.[58] On the basis of literature surveys, it is clear that the key for direct conversion of HMF to 1,6-HDO demands dual functionality for C=C and C−O cleavages; thus, acidic promoters are preferable for rational design of good catalysts. The plausible reaction pathways on various solid catalysts can be seen in Figure 11.

Adipic Acid from Lignin. Lignin and derivatives have also been widely exploited for potential production of adipic acid. To date, biological conversion routes of lignin derivatives starting from catechol or phenol provide promising opportunities in this context. Oxidative ring-opening reactions catalyzed by biocatalysts lead to muconate as a key intermediate for the synthesis of adipic acid from lignin (Figure 12a).[39,185,186] On the other hand, catalytic hydrogenations of substituted phenols such as guaiacol to cyclohexanol and cyclohexanone represent alternative routes that could potentially revolutionize the current nylon industry. This is because conversion of phenol derivatives only demands hydrogenation catalysts that facilitate saturation of C=C bond in aromatics, while catalytic cascade reactions involving oxidation of so-called lignin derived "KA oil" easily give adipic acid as the major product (Figure 12b).[5]

Different from hydrogenation of benzene, toluene, and xylene, saturation of phenols demands selective catalysts that will only elegantly promote C=C cleavage. There have been several successful attempts in transforming phenol and guaiacol into relevant cyclic alcohols and aldehyde using noble Pt and Pd catalysts, even under very mild conditions.[5,64] Metal−support interactions play key roles in selective activation of C=C bonds without interrupting C−O bonds in phenol rings. Lewis acid sites are strongly believed to favor formation of cyclohexanone via the heterogenic split of H_2 molecules due to the asymmetrical structure of phenol. Bimetallic catalysts could also enhance activity and selectivity, while very limited work is yet to be published in the literature.

However, one may ignore the fact that depolymerization of lignin, or purification of bio-oil, should be the bottlenecking step for economical synthesis of adipic acid from lignin derivatives. Impurities and toxic compounds, such as organic sulfurs and inorganic salts in crude biomass, easily deactivate metal catalysts.[187−191] Those unwanted impurities often interact strongly with solid catalysts and form stable and inactive compounds, which display poor catalytic activity for oxidation and hydrogenation reactions. Therefore, more efforts should be paid to effective separation of key lignin-derived intermediates for catalytic cascade reactions for adipic acid production.

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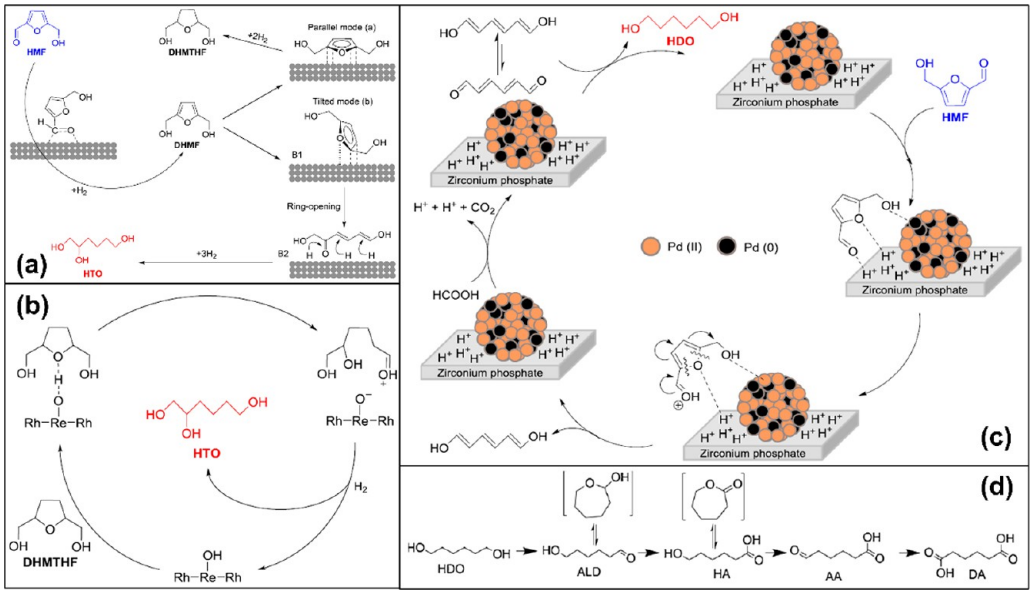


Figure 11. Detailed reaction pathways for conversion of HMF and derivatives over different catalysts: (a) hydrogenation of HMF to 1,2,6-hexanetriol on CoNi catalyst (permission from ref 181), (b) hydrogenation of 2,5-dihydroxymethyltetrahydrofuran over RhRe catalyst,[182] (c) conversion of HMF to 1,6-HDO on Pd/ZrP catalyst (permission from ref 183), and (d) oxidation of 1,6-HDO to adipic acid (permission from ref 184).

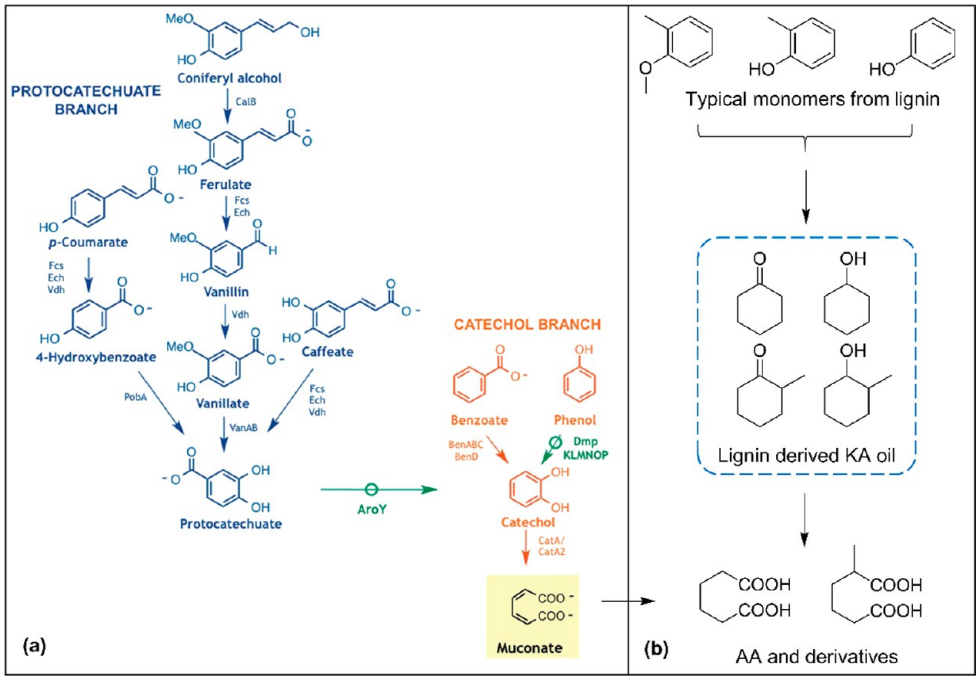


Figure 12. Adipic acid production from lignin via (a) biological route with muconate as key intermediate[6,185,186] and (b) proposed chemical route with cyclohexanol derivatives as key intermediates.

CONCLUSIONS AND PERSPECTIVES

The chemistry and catalyst designs for facile adipic acid synthesis from biomass have been critically discussed and revised. Detailed conversion pathways for glucose-derived adipic acid and the role of metal−metal and metal−support interaction on activity, selectivity, and durability of nanostructured catalysts suggests that the formulation of nanostructured materials play a key role for tunable C−H and C−O bond cleavage/formation in bioderived intermediates. With the primary focus on (a) morphologies of nanoparticles, (b) electronic coupling behaviors in hybridized nanocomposites, and (c) confinement of porous supports, it is further revealed that the morphological and electron coupling effects in bimetallic and metal−metal oxide catalysts are determining factors for sustainable production of adipic acid from lignocellulose.

It is important to point out that the overall atom efficiencies for the FDCA and glucaric acid routes are similar, as both routes require oxidation of terminal −C−OH and −CHO groups, with either C−O cleavage or C═O bond saturation for four internal carbons. However, based on the conversion and selectivity data reported in the literature, the FDCA route displays relatively better atom efficiency, as conversion of glucose and fructose to HMF and subsequent oxidation of HMF to FDCA show

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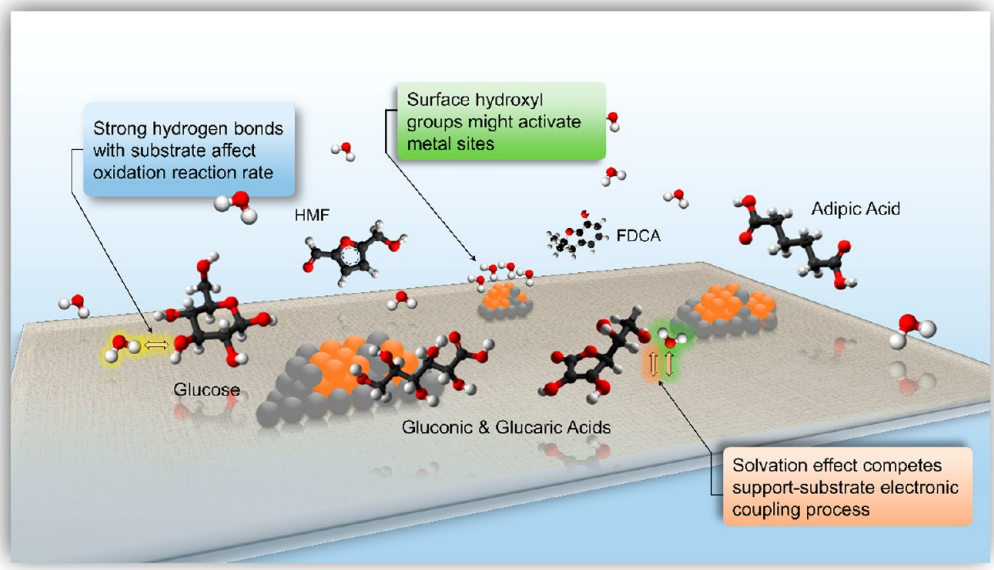


Figure 13. Key unsolved issues for selective adipic acid synthesis from glucose over nanocatalysts.

relatively higher chemoselectivity compared with glucose oxidation to glucaric acid. According to our literature survey, it is clear that HMF conversion to FDCA is fairly easy to achieve in the presence of solid catalysts (99+ % yield). But oxidation of glucose to glucaric acid is still very difficult to achieve, as the best results were reported on Pt/C catalysts with less than 80% yield. In addition, hydrogenolysis of FDCA to adipic acid even occurs in the absence of solid catalysts, while deoxygenation of glucaric acid is still under fundamental research using homogeneous (HI/HBr) and expensive heterogeneous noble metal catalysts. Considering the increasing awareness of FDCA in replacing PET plastics in the chemical industry, the price for FDCA might further lower owing to the technological development of oxidation processes.

The most advantageous aspect of the glucaric acid route is that direct oxidation of glucose needs only one-pot conversion to form the target product. Thus, it shows overall high atom efficiency compared with multistep conversion of glucose to FDCA. The lack of studies on separation and purification techniques, owing to complicated product mixtures with high boiling points, is a major limiting factor in fundamental and practical efforts. Hydrogenolysis of glucaric acid, similar to the case of FDCA, occurs fairly easily in the presence of metal catalysts and HBr/HI promoters.

There is still a large gap between quantitative synthesis in lab scale and successful implementation of bioadipic acid technology in industrial scale. Due to poor intrinsic selectivity for activation of C−H and C−O bonds, several key issues still remain to be addressed in this area. In particular, the following two aspects are still largely unexplored (illustrated in Figure 13):

(1) Solvent or cosolvent effect for tunable product selectivity. Although water is known to be poor solvent for both oxidation and hydrogenation reactions, it seems to be necessary in this area, as water often coexists in large quantity with sugars and lignin compounds.[192,193] Earlier studies on the solvent effect have been focused on the critical role of aprotic solvents on the stabilization of acidic protons. Most experimental and computational efforts have been on revealing the mechanism of dehydration reactions. The key finding is that polar aprotic solvents affect the stabilization process of the acidic proton, leading to enhanced reaction rates for the acid-catalyzed biomass conversion reaction. Information on the molecular interaction of polar (as well as nonpolar) solvents with sugar molecules and catalyst surfaces is still unrevealed in this area. While it is well known that electronic coupling effects induced a strong interaction between −O−H, and a metal oxide surface often leads to enhanced Lewis acidity of solid catalytic materials. Low but detectable oxidative environments in CO_2 expanded media are also known to facilitate the C−H bond.[194−197] These promising features have yet to be investigated for facile oxidation and a hydrogenolysis reaction involving C−O and C═O bond activation.

In addition to their unique promotional effects on oxidation, the impact of catalyst stability on solid supports should be studied in detail. Although numerous studies have shown good performances of heterogeneous catalysts in aqueous media, the sintering of active sites and decomposition of solid supports should be particularly investigated in future studies.

(2) Adsorption-induced lattice distortion on catalyst surface. As a classic topic in environmental and energy catalysis involving small molecules, adsorption is known to induce surface reconstruction of nanosized catalytic materials. The fact that current bioadipic acid is carried out in a liquid phase prevents researchers from obtaining insightful yet reliable information at operando conditions. However, no one would be surprised to find that the adsorption of relatively large sugar molecules (compared with H_2, O_2, light hydrocarbons, etc.) alters local structures of active sites on catalyst surfaces. With the insights provided by computational studies, future work should be more focused on truly rational design of catalytic materials which show both active and durable performances in aqueous reaction media.

Other catalytic approaches, such as electrochemical and photo-oxidation of glucose to glucaric acid, can provide a means to achieve oxidation mildly compared to conventional thermal

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treatment and shows promising performances.[126,198,199] Literature reported electro-oxidation of glucose to glucaric acid in the presence of sodium hypochlorite, sodium bromide, and TEMPO and achieved over 90% yield. However, the requirements of these reagents generate toxic byproducts, and their stoichiometric addition makes the ECO process economically unappealing on a large scale.

Other engineering aspects, such as multiphase mass transformation limitation in gas−liquid−solid reaction systems and flow dynamic behaviors in continuous reactors, are also needed to be well studied by collaborative work among chemists and engineers.[200] This lies at the foundation of successful application of the concepts of bioadipic acid in future biorefineries.

In all, synthesis of renewable adipic acid opens a new opportunity for the chemical industry that steps into a more sustainable future. The synergy between chemistry, surface science, and catalysis undoubtedly will play an increasingly critical role for sustainable development of the human society.

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X.J. proposed the topic and drafted the manuscript. M.L., G.Z., J.W., Q.X., Y.S., Z.Z., W.Z., and S.W. assisted in the literature search and material collection. C.Y., J.S., and C.H.L. discussed and revised the manuscript.

Notes

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